

Multimodal Learning for Autoencoders

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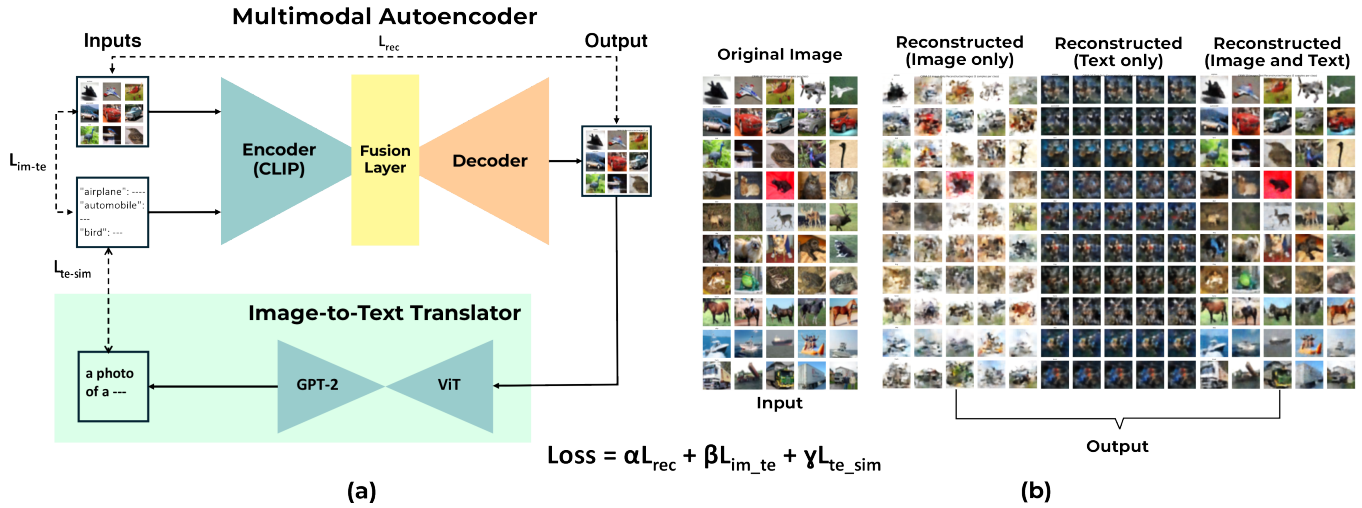


Figure 1: (a): Model architecture and training scheme including Multimodal Autoencoder, Image-to-Text Translator, pre-trained models (CLIP, ViT, GPT-2) and the loss functions L_{rec} , L_{im-te} , L_{te-sim} . (b): Original Images from CIFAR-10 dataset (b1) and their reconstructed outputs from image only input (b2), text only input (b3), image and text combined inputs (b4).

Abstract

In this work, Multimodal Autoencoder is proposed in which images are reconstructed using both image and text inputs, rather than just images. Two new loss terms are introduced: Image-Text-loss (L_{im-te}) that measures similarity between input image and input text, and Text-Similarity-loss (L_{te-sim}) that measures similarity between generated text from the reconstructed image and input text. Our experiments demonstrate that images reconstructed using both modalities (image and text) are of significantly higher quality than those reconstructed from either modality alone, highlighting how features learned from one modality can enhance another in the reconstruction process.

CCS Concepts

• **Computing methodologies** → **Reconstruction**; *Image representations*; **Neural networks**; *Information extraction*.

Keywords

Computer Vision, Natural Language, Autoencoder, Multimodality,

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1 Introduction

Autoencoders have been typically used to generate images by reconstructing the original image which only involves one modality. However, our research explores the inclusion of additional modality such as text, to train a Multimodal Autoencoder. While some work has already been done in this domain [Jaques et al. 2017] [Geng et al. 2022], our approach employs a novel architecture and training scheme. Modeling a multimodal autoencoder presents several challenges, like the need for well-represented and aligned modalities in a shared latent space and the efficient and correct transference of knowledge between different modalities so that it can help in the reconstruction of the input image [Liang et al. 2023]. We address these challenges by utilizing OpenAI's CLIP [Radford et al. 2021] and a Vision Transformer [Dosovitskiy et al. 2021], GPT-2 [Radford et al. 2019] based Image-to-Text translator. CLIP uses a contrastive learning approach rather than a predictive one and works very well for handling images and text. We incorporate pre-trained CLIP with frozen parameters and train our own decoder. This approach necessitates the use of our own input images and their respective input texts, with the model's training influenced by how we craft these input texts. After generating features for our image and text, we fuse them and pass them to the decoder, which reconstructs an

image based on these features. To further refine the text modality, we introduce a pre-trained Image-to-Text translator which in our case is ViT and GPT-2 based Encoder Decoder architecture that generates text for our reconstructed image, which is then compared with the input text. This multi-faceted and multimodal approach allows our model to reconstruct images that are impacted by both the input image and the input text.

2 Method

Our model consists of two main parts, first part is Multimodal Autoencoder, which consists of CLIP Encoder, Fusion Layer and Decoder. It is responsible for image reconstruction from multimodal inputs. Second part is generated image to input text translator, which we use to calculate text similarity between translated text and input text providing reconstruction loss calculation between different modalities. We use pre-trained CLIP[Radford et al. 2021] model ("openai/clip-vit-base-patch32") from the transformers library of hugging face for both image and text encoding. CLIP's image encoder generates image features, while its text encoder processes associated text. Then these features are concatenated in the fusion layer of our Multimodal Autoencoder, which reduces the concatenated CLIP image and text features to the CLIP dimension (512). These features are then passed to a decoder which consists a series of transposed convolution layers that upscale the fused features to generate a 224x224x3 image. This generated image is then passed to Image-to-Text translator, which then translates the image back to text so that we can compare it to the input text. We use a pre-trained VisionEncoderDecoderModel ("nlpconnect/vit-gpt2-image-captioning") from the transformers library of hugging face for translating generated images back to text. This model combines a ViT (VisionTransformer)[Dosovitskiy et al. 2021] encoder with a GPT-2 decoder[Radford et al. 2019]. Our primary goal is to reconstruct the original image, so we employ reconstruction loss (L_{rec}). However, with two modalities involved, we introduce an additional loss term between the input image and input text, which we call Image-Text-Loss (L_{im-te}). This ensures that our image and text are aligned together. To further refine the text modality, we introduce Image-to-Text translator that generates text for our generated image. This generated text is then compared with the input text to calculate Text-Similarity-Loss (L_{te-sim}). Final loss function is the combination of three components: L_{rec} , mean squared error between input image and reconstructed image. L_{im-te} , cosine similarity between input image and input text. L_{te-sim} , cosine similarity between input text and translated text. Final loss is the weighted sum of all of these components:

$$Loss = \alpha L_{rec} + \beta L_{im-te} + \gamma L_{te-sim}$$

The values of α , β and γ are empirically set to 1, 0.2 and 0.3 respectively in our evaluation.

3 Experimental Evaluation and Discussion

We implemented our model using the CIFAR-10 dataset, which contains 60,000 32x32 color images across 10 classes. The images were resized to 224x224 pixels and normalized. Each image was paired with a detailed textual description of its class. Rather than using generic class descriptions (e.g., "a photo of an airplane"), we

crafted descriptions that our Image-Text Translator could effectively capture. We found that the translator performs well with terms related to objects, colors, and backgrounds, so we incorporated these elements into our textual descriptions (e.g., "a photo of a large white airplane flying in the clear sky."). This approach allows the model to learn stronger associations between the input text and input image. We freeze both CLIP and Image-to-Text translator models, as they have already been extensively trained. Our model was trained for 300 epochs. Post-training, we evaluated the model using various metrics, with a primary focus on image reconstruction quality, given our goal of designing a Multimodal Autoencoder. As shown in Figure 1 (b), reconstructing images from image only input (b2) or text only input (b3) results in lower quality reconstructions. Image-only input (b2) produces better results than text-only input (b3), which is expected given the higher weight of L_{rec} in the final loss term compared to L_{te-sim} . Importantly, when we reconstruct images using both image and text inputs (b4)—incorporating two modalities—the final reconstructed image quality improves significantly. This demonstrates that our Autoencoder is truly Multimodal, learning to reconstruct images not just from image alone, but also from text. By examining the results of the image and text reconstructed images, we can observe that the model is learning to fill in missing information from both modalities individually and is able to leverage this combined information for the final reconstruction. This suggests that our multimodal approach is successfully integrating information from both the visual and textual domains to produce higher-quality image reconstructions.

4 Conclusion

We introduce a novel Multimodal Autoencoder architecture and training scheme that effectively handles both image and text inputs for improved image reconstruction. This work can be further extended to include different modalities like audio and different kind of signals, both as an input and as an output.

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